Write a python program for transposition ciphers using rail fence technique and columnar technique

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Cryptography

- Cryptography comes from the Greek words for "*secret writing*".
- **Cipher** is a character-for-character or bit-for-bit transformation
  - No regard to the linguistic structure of the message
- **Code** replaces one word with another word or symbol.

Symmetric-Key ciphers

- A **symmetric-key** cipher uses the same key for both encryption and decryption
  - Key can be used for bidirectional communication
- Traditional symmetric-key Ciphers are classified as
  - **Substitution ciphers**
  - **Transposition ciphers**

Encryption Model

- Encryption of the plaintext P using key K gives the ciphertext C

$$C = E_K(P)$$

- $P = D_K(C)$ represents the decryption of C to get the plaintext again.

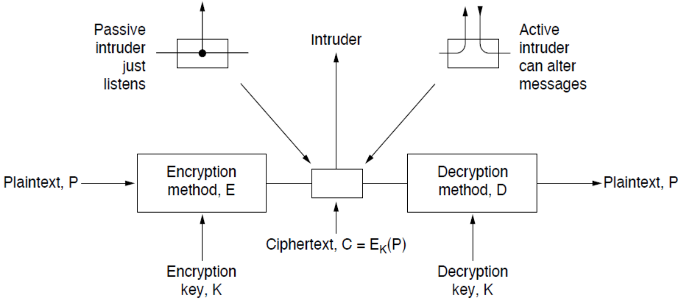

Figure: Encryption model of a Symmetric-key cipher

Substitution Cipher

- In a substitution cipher, each letter (group) of letter(s) is replaced by another letter(s) to disguise it.
- **Caesar cipher**: $a \implies D$, $b \implies E$,...,$z \implies C$
  - $attack \implies DWWDFN$
- If the symbols are digits (0 to 9)
  - We can replace $3 \implies 7$ and $2 \implies 6$.
- **Generalization of the Caesar cipher**: Alphabet to be shifted by k letters, instead of three

Transposition Ciphers

- A transposition cipher reorders symbols
- **Rail Fence Cipher**: Encryption
  - **Step 1**: The plain text is written as a sequence of diagonals.
  - **Step 2**: To obtain the cipher text, the text is read as a sequence of rows
- Example : "INCLUDEHELPISAWESOME". Key=2
- No of columns=length of the text=20; No Of Rows=2 (Key)

I		C		U		E		E		P		S		W		S		M	
	N		L		D		H		L		I		A		E		O		E

- Write the message in a zigzag manner then read it out direct row-wise to change it to cipher-text
- Cipher-text: ICUEEPSWSMNLDHLIAEOE
- Rail-Fence Technique is very easy to break by any cryptanalyst.

Rail Fence Cipher: Encryption

Examples: `https://www.geeksforgeeks.org/rail-fence-cipher-encryption-decryption/`

- Input : "attack at once"; Key=2;
- No of columns=length of the text=14; No Of Rows=2 (Key)

a		t		c		-		t		o		c	
	t		a		k		a		-		n		e

- Output : atc toctaka ne

- Input : "defend the east wall"; Key=3;
- No of columns=length of the text=20; No Of Rows=3

d				n				h				a				w			
	e		e		d		t		e		e		s		-		a		l
		f				-				-				t				l	

- Output : dnhaweedtees alf tl

Rail Fence Cipher: Decryption

- Input : "atc toctaka ne" ; Key=2;
- No Of Columns=14; Rows=2; write "*" in a zigzag manner

*		*		*		*		*		*		*	
	*		*		*		*		*		*		*

- Replace "*" with the characters of the cipher text

Rail Fence Cipher: Decryption

- Input : "atc toctaka ne" ; Key=2;
- No Of Columns=14; Rows=2; write "*" in a zigzag manner

*		*		*		*		*		*		*	
	*		*		*		*		*		*		*

- Replace "*" with the characters of the cipher text

a		t		c		-		t		o		c	
	t		a		k		a		-		n		e

- Output : "attack at once"

Columnar Transposition Technique

Steps to obtain cipher text

- In a rectangle of pre-defined size, write the plain-text message row by row.
- Read the plain message in random order in a column-wise fashion. It can be any order such as 2,1,3 etc.
- Cipher-text is obtained!!

Columnar Transposition: Encryption Example

Original message: "INCLUDEHELPISAWESOME".

C1	C2	C3	C4
I	N	C	L
U	D	E	H
E	L	P	I
S	A	W	E
S	O	M	E

4	2	3	1
L	N	C	I
H	D	E	U
I	L	P	E
E	A	W	S
E	O	M	S

- column key="VEGA"
- Alphabetical order in the given key "VEGA" $\Longrightarrow$ 4231
- **Plaintext is written horizontally, in rows**
- Reorder the columns
- Ciphertext is read out by columns: LHIEENDLAOCEPWMIUESS

Columnar Transposition: Decryption

Given: LHIEENDLAOCEPWMIUESS; column key = "VEGA" $\Longrightarrow$ 4231

Write the text in 4 columns (column-wise)

4	2	3	1
L	N	C	I
H	D	E	U
I	L	P	E
E	A	W	S
E	O	M	S

Reorder the columns as per key

1	2	3	4
I	N	C	L
U	D	E	H
E	L	P	I
S	A	W	E
S	O	M	E

Read the text row-wise:
"INCLUDEHELPISAWESOME"